UE23CS352B - Object Oriented Analysis & Design

Mini Project Report

Movie Ticket Booking System

Submitted by:

Team Members

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Semester VI - Section 'I'

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Problem Statement:

Traditional movie ticket booking processes can be inefficient and difficult to manage, especially when handling multiple operations such as user authentication, movie selection, and ticket booking. Without a structured design, the system can become complex, hard to maintain, and tightly coupled.

This project aims to develop a scalable and maintainable online cinema booking system by applying design patterns and SOLID principles to simplify the booking workflow and improve system organization.

Key Features:

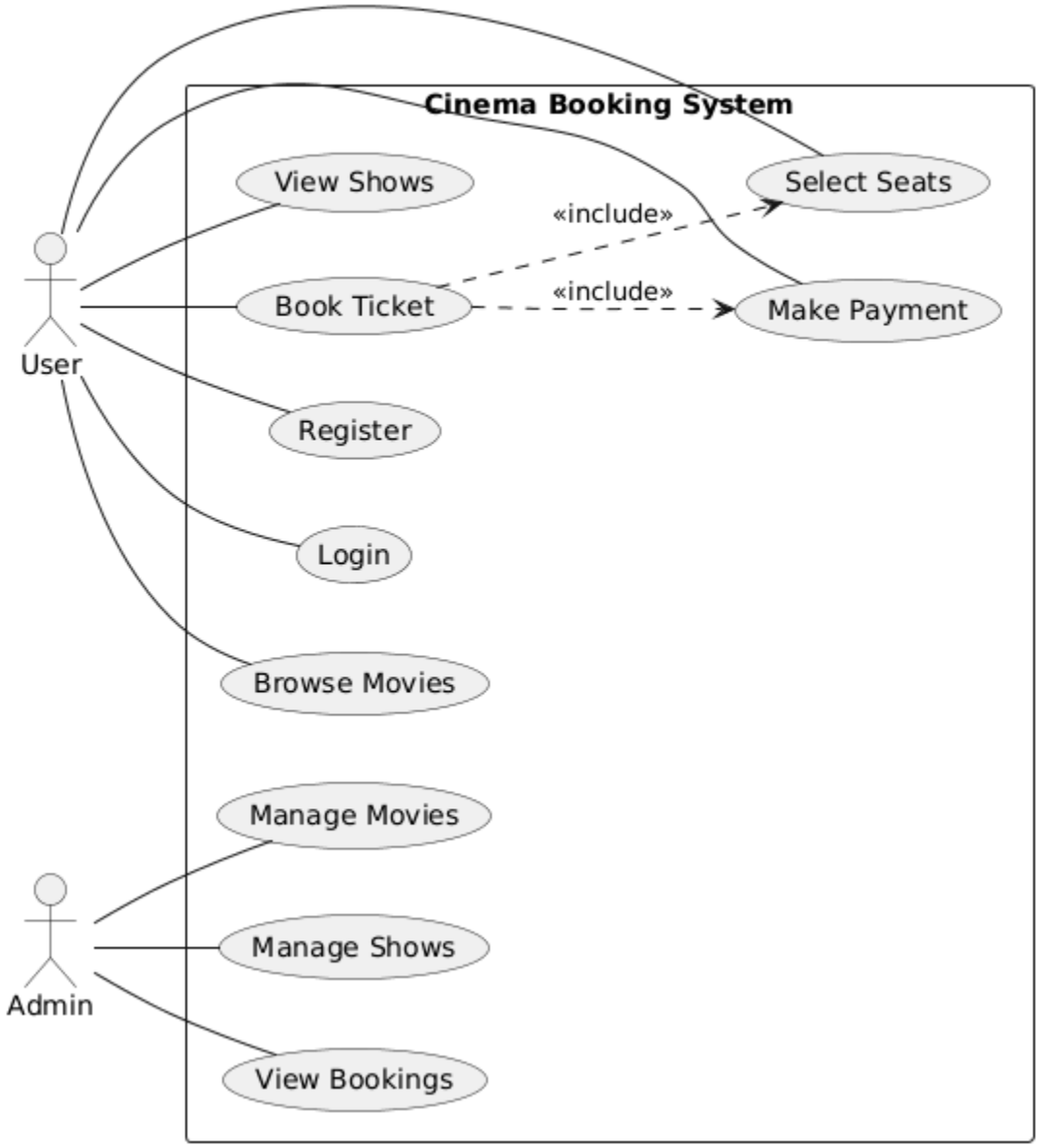
- User registration and login system
- Browse available movies and show timings
- Ticket booking with simplified booking flow
- Booking history and confirmation display
- Structured backend using design patterns and SOLID principles
- Modular and maintainable code architecture
- Database integration for storing users, movies, and bookings

Models:

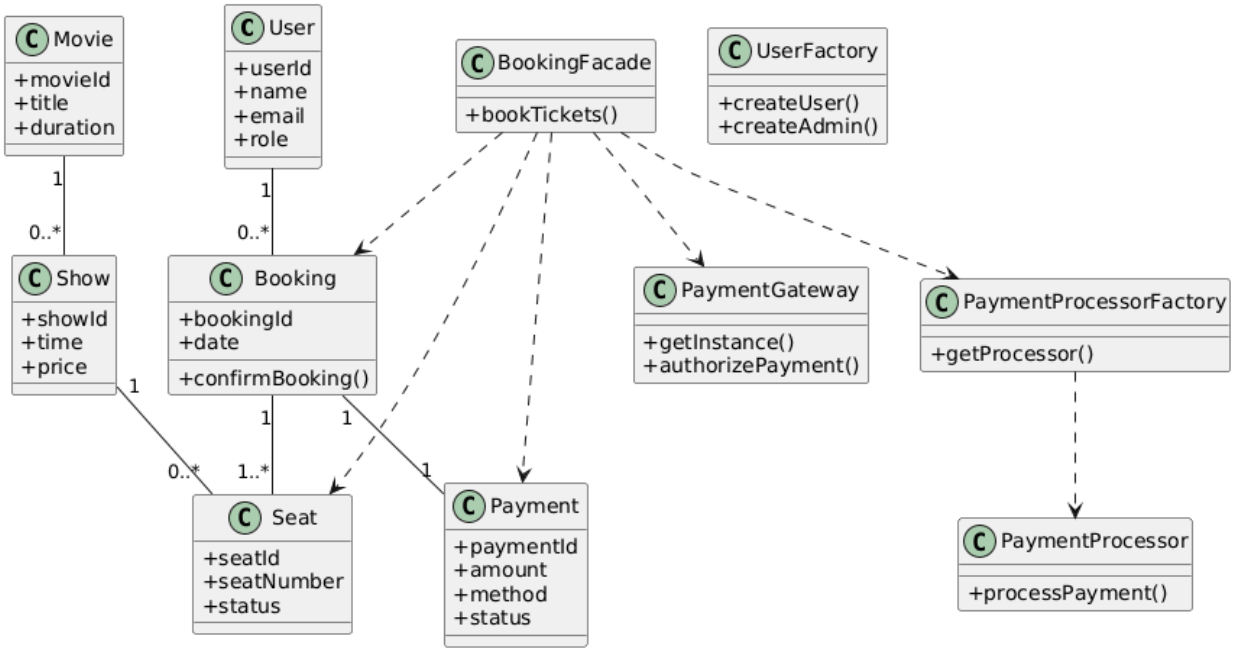
Entities:
User, Movie, Show, Booking, Payment

Enums:
UserRole (ADMIN, USER),
PaymentType (CARD, UPI),
BookingStatus (CONFIRMED, CANCELLED),
ShowTiming

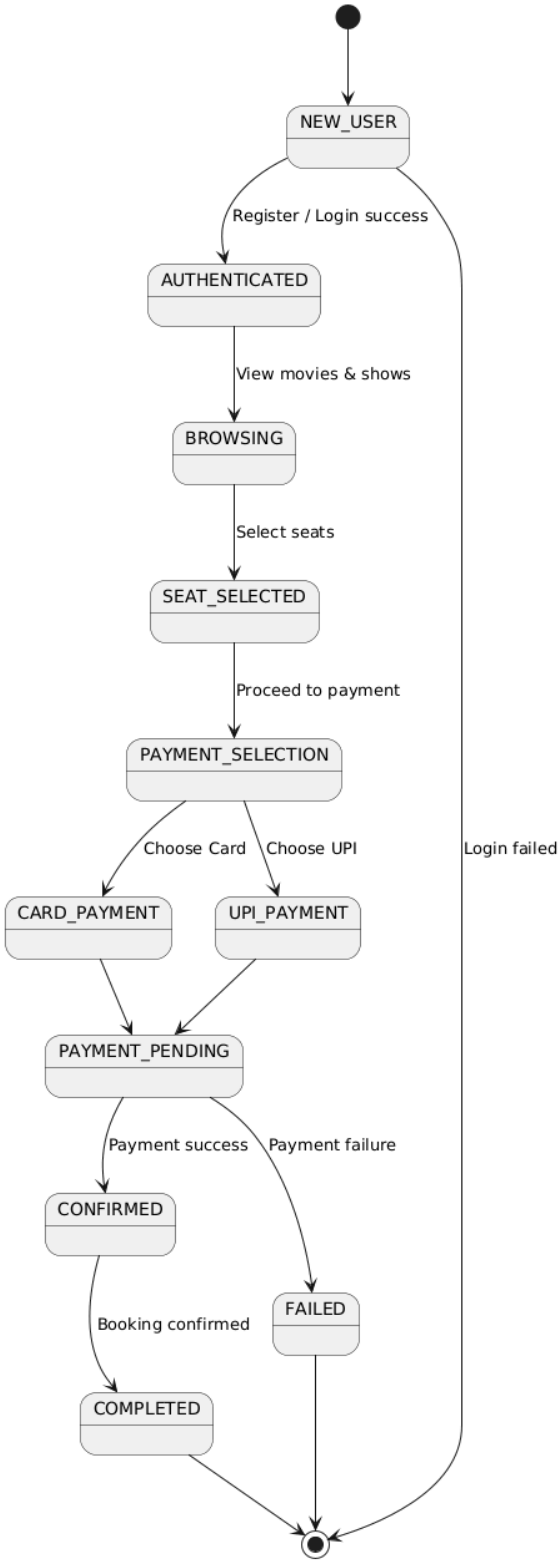
Use Case Diagram:



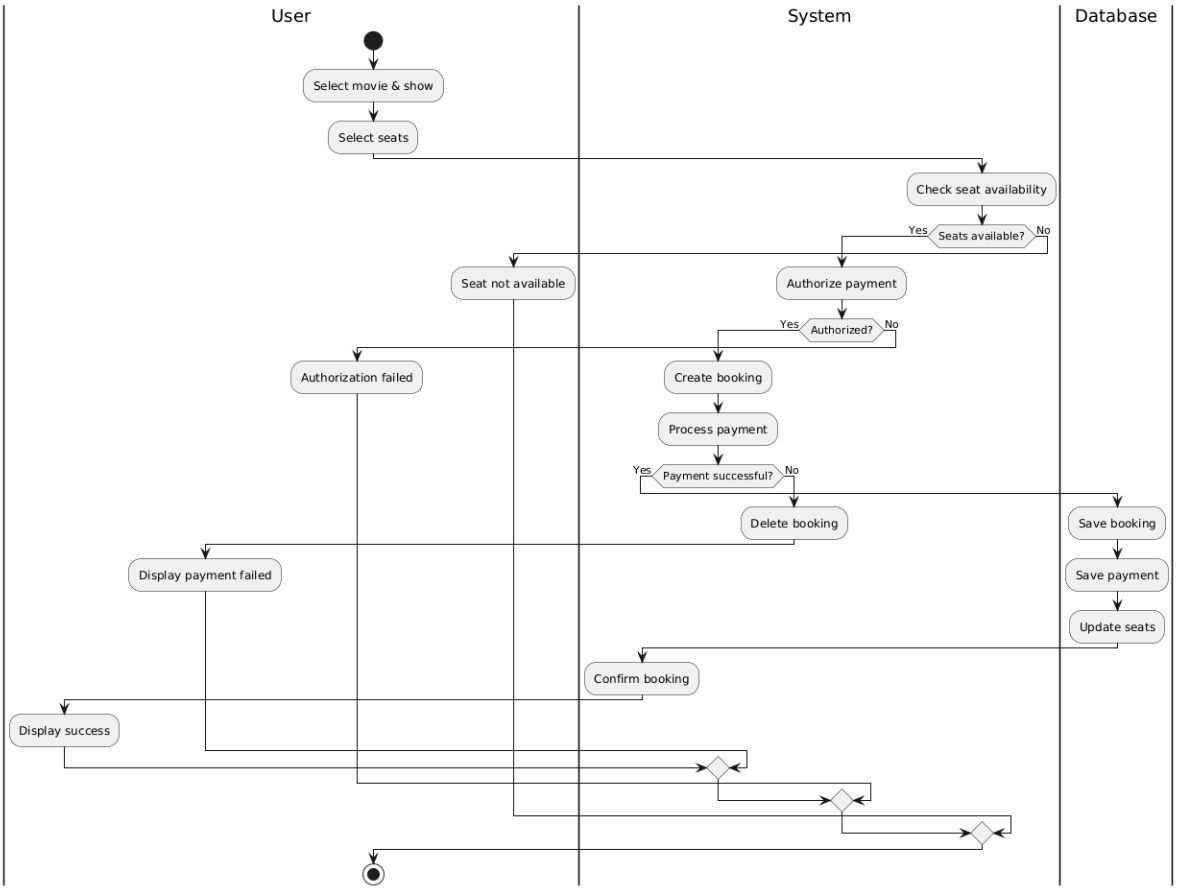
Class Diagram:



State Diagram:



Activity Diagram:



Design Principles, and Design Patterns:

<u>MVC Architecture used? -</u> Yes

The project follows the **Model-View-Controller (MVC)** architecture -

- **Model:**
  Represents data and business logic
  → Classes like `User`, `Movie`, `Booking`, and repositories handle data storage and operations
- **View:**
  Handles the user interface
  → Thymeleaf templates display movies, bookings, login pages, etc.
- **Controller:**
  Manages user requests and responses
  → Classes like `BookingController` and `AuthController` process inputs and interact with the model

<u>Design Principles</u>

- **Single Responsibility Principle (SRP):**
  Each class has only one job (e.g., controller handles requests, repository handles DB).

- **Open/Closed Principle (OCP):**
  Code is open for extension but closed for modification (e.g., new user types via factory without changing existing code).

- **Liskov Substitution Principle (LSP):**
  Subclasses can replace parent classes without breaking functionality.

- **Dependency Inversion Principle (DIP):**
  Depend on abstractions, not concrete implementations (e.g., using interfaces for services/repositories).

Design Patterns

- **Factory Pattern:**
  Used for creating user objects (`UserFactory`) without exposing creation logic.

- **Facade Pattern:**
  Simplifies the booking process by providing a single interface (`BookingFacade`) to handle multiple operations.

- **Singleton Pattern:**
  Ensures only one instance of the payment gateway exists (`PaymentGateway`) for centralized control.

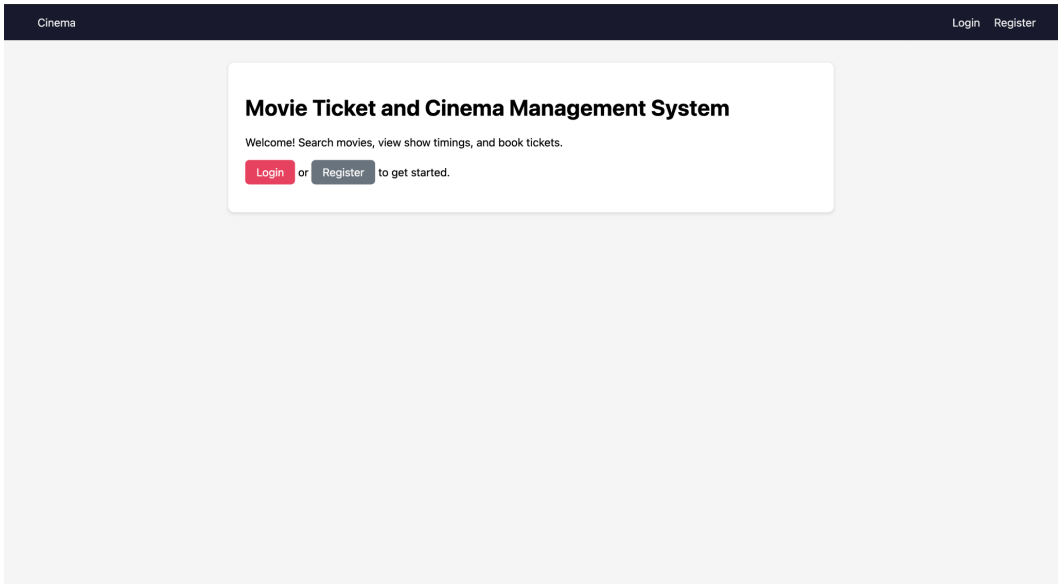
- **Builder Pattern:**
  Used to construct complex booking objects step-by-step (`BookingBuilder`).

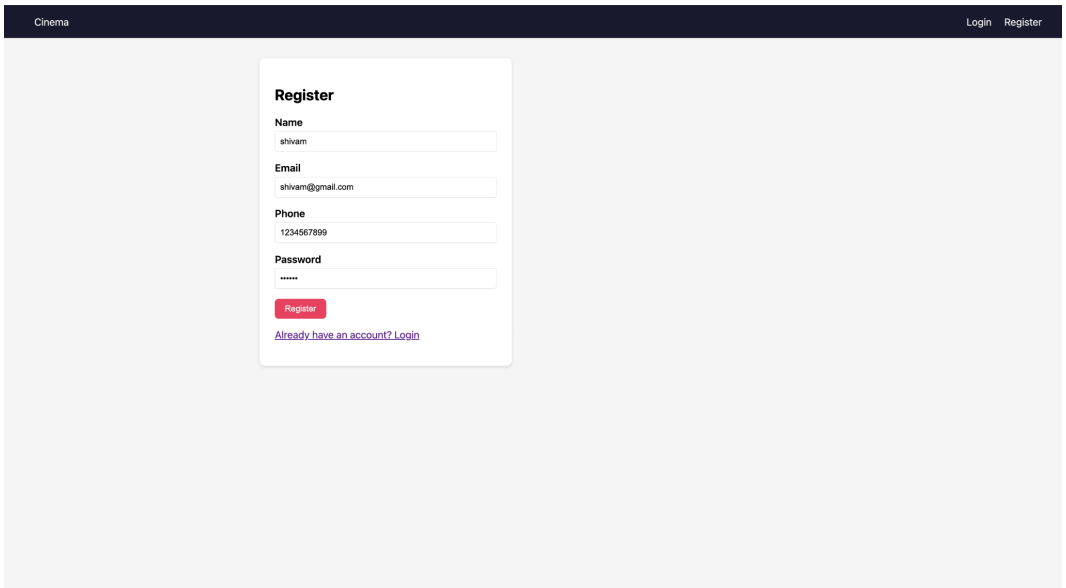
Github link to the Codebase: (repository should be public)

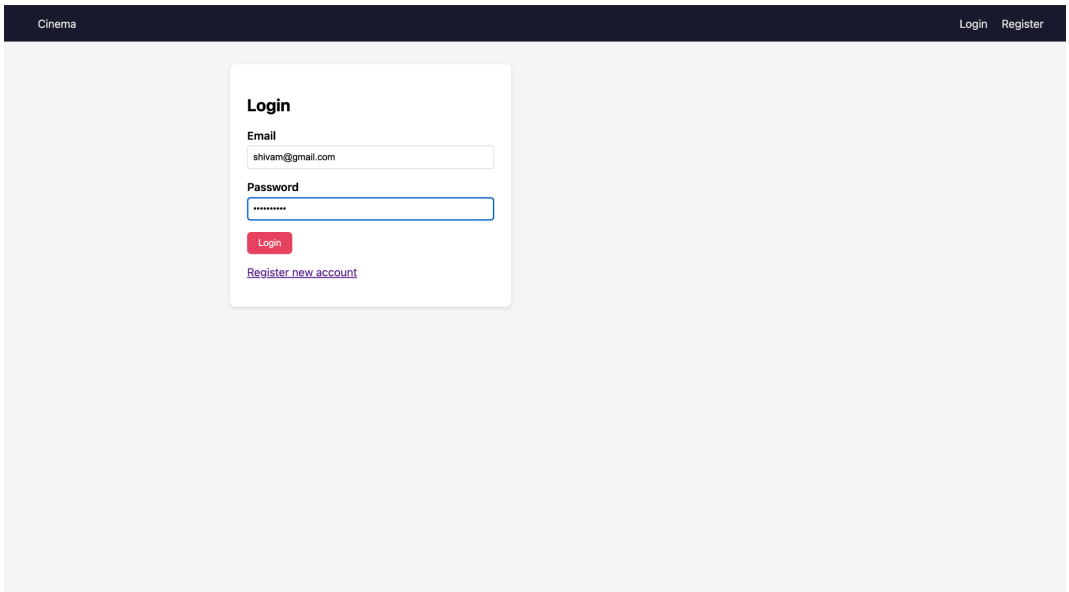
https://github.com/sshivamanand/movie_ticket_booking_system

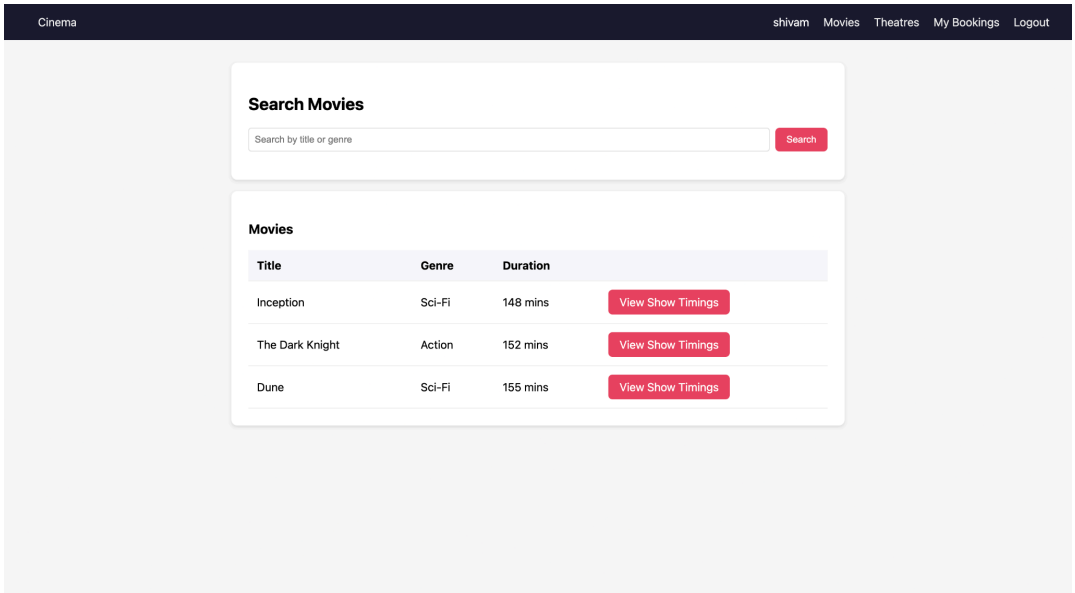
Screenshots

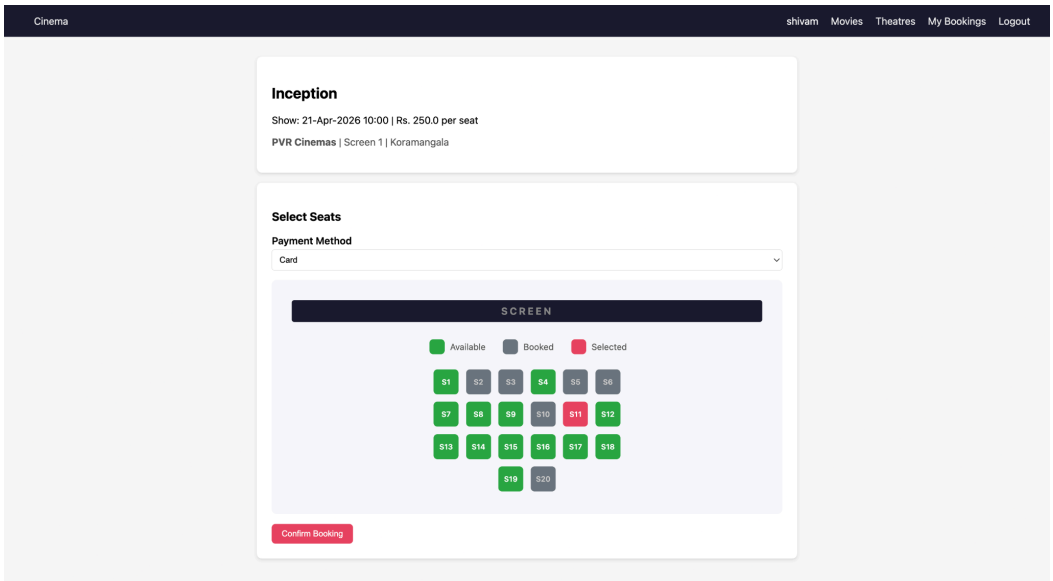
UI:

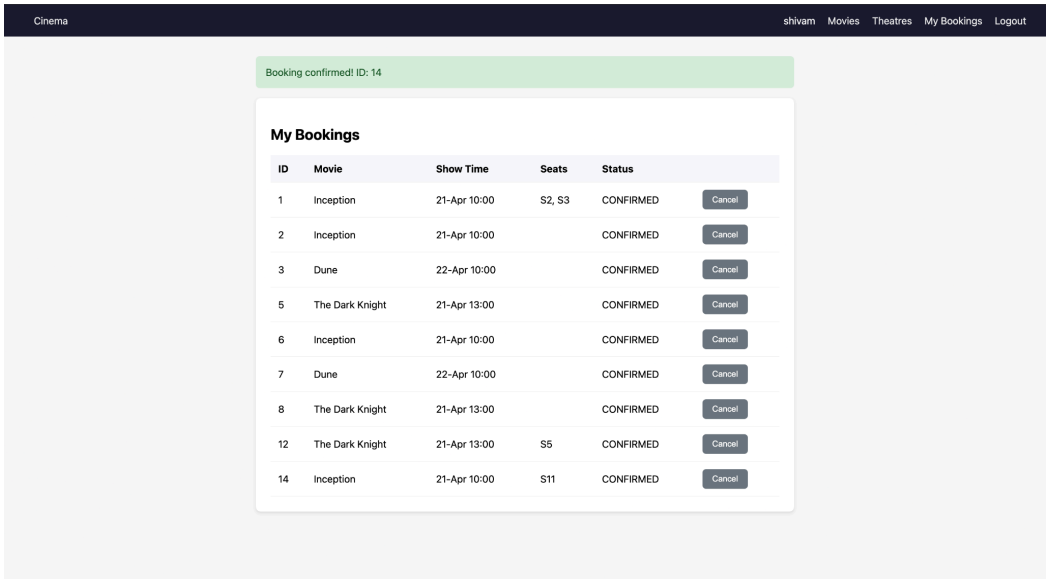





Cinema
Login
Register
Login
Email
shivam@gmail.com
Password
Login
Register new account


Cinema
shivam
Movies
Theatres
My Bookings
Logout
Search Movies
Search by title or genre
Search
Movies
Title
Genre
Duration
Inception
Sci-Fi
148 mins
View Show Timings
The Dark Knight
Action
152 mins
View Show Timings
Dune
Sci-Fi
155 mins
View Show Timings


Cinema
shivam Movies Theatres My Bookings Logout
Inception
Show: 21-Apr-2026 10:00 | Rs. 250.0 per seat
PVR Cinemas | Screen 1 | Koramangala
Select Seats
Payment Method
Card
SCREEN
Available Booked Selected
S1 S2 S3 S4 S5 S6
S7 S8 S9 S10 S11 S12
S13 S14 S15 S16 S17 S18
S19 S20
Confirm Booking


Cinema
shivam Movies Theatres My Bookings Logout
Booking confirmed! ID: 14
My Bookings
ID Movie Show Time Seats Status
1 Inception 21-Apr 10:00 S2, S3 CONFIRMED Cancel
2 Inception 21-Apr 10:00 CONFIRMED Cancel
3 Dune 22-Apr 10:00 CONFIRMED Cancel
5 The Dark Knight 21-Apr 13:00 CONFIRMED Cancel
6 Inception 21-Apr 10:00 CONFIRMED Cancel
7 Dune 22-Apr 10:00 CONFIRMED Cancel
8 The Dark Knight 21-Apr 13:00 CONFIRMED Cancel
12 The Dark Knight 21-Apr 13:00 S5 CONFIRMED Cancel
14 Inception 21-Apr 10:00 S11 CONFIRMED Cancel

Individual contributions of the team members:

Name	**Module worked on**
SHASHWAT SOLANKI	Payment Module
SHAUNAK A. RAI	Booking Construction Module
SHIVAM ANAND	Booking Module
SHREYAS SREENIVAS	User Module